MSc Data Science
DS601: Mathematical Foundations for Data Science
Date: 24/08/2026

Tutorial 4

This tutorial is divided into two sections. In Section A, solve the questions analytically. In Section B, implement the same problems computationally in Python, increase the scale of the problems, and study how the numerical results change.

Section A: Analytical Questions

Q1) Binomial Distribution

Suppose an experiment is performed 4 independent times. In each experiment, there are only two possible outcomes: Success or Failure.

The probability of success in each experiment is:

$$\mathrm{P(Success)} = \mathrm{p} = 0.85$$

The probability of failure in each experiment is:

$$\mathrm{P(Failure)} = \mathrm{q} = 0.15$$

Let the random variable X represent the number of successes obtained in the 4 experiments. Therefore:

$$\mathrm{X} \sim \mathrm{Binomial}(\mathrm{n} = 4, \mathrm{p} = 0.85)$$

Answer the following questions analytically:

(a) Write the probability mass function (PMF) of a Binomial distribution for obtaining exactly x successes in n independent experiments.

(b) For n = 4, write all possible values that the random variable X can take.

(c) Calculate the probability of obtaining exactly 0, 1, 2, 3, and 4 successes. Organize your calculations in the following table:

Number of Successes, x	P(X = x)
0	
1	
2	
3	
4	

(d) Add the probabilities of all possible outcomes:

$$\mathrm{P(X = 0)} + \mathrm{P(X = 1)} + \mathrm{P(X = 2)} + \mathrm{P(X = 3)} + \mathrm{P(X = 4)}$$

What value do you obtain? Explain why the sum of the probabilities of all possible outcomes of a discrete probability distribution must be equal to 1.

(e) From the values calculated above, determine the maximum probability.

(f) At what value of x does this maximum probability occur? Hence, determine the most likely number of successes in 4 experiments.

(g) The probability of success in an individual experiment is 0.85. Based on your calculated distribution, explain why outcomes containing a larger number of successes are more likely than outcomes containing a smaller number of successes.

Q2) Continuous Probability Distribution

Consider a continuous random variable X having the following probability density function:

$$f(x) = 6x(1 - x), \quad 0 \leq x \leq 1$$

and f(x) = 0 outside this range.

For this question, take:

$$\Delta x = 0.1$$

Therefore, consider the points:

$$x = 0, 0.1, 0.2, 0.3, ..., 0.9, 1.0$$

Answer the following questions analytically:

(a) Calculate f(x) at each of the given values of x. Organize your calculations in the following table:

x	f(x) = 6x(1 - x)
0.0	
0.1	
0.2	
0.3	
0.4	
0.5	
0.6	
0.7	
0.8	
0.9	
1.0	

(b) From the calculated values, find (i) the maximum value of f(x), and (ii) the value of x at which this maximum occurs.

(c) Using the calculated values, approximate the total probability using:

$$S = \boldsymbol{\Sigma}_i \, f(x_i) \, \Delta x$$

Use Δx = 0.1. Is the calculated value exactly equal to 1?

(d) Compare the following two quantities:

$$\boldsymbol{\Sigma}_i \, f(x_i)$$

and

$$\boldsymbol{\Sigma}_i \, f(x_i) \, \Delta x$$

Which one represents an approximation of the total probability? Explain why the values of f(x) alone should not simply be added and expected to give 1.

(e) For a continuous random variable, what is P(X = 0.5)? Explain briefly the difference between probability density f(x) and probability.

Section B: Computational Questions

Q3) Computational Analysis of the Binomial Distribution

Write a Python program to reproduce and extend the Binomial distribution studied in Q1. Use p = 0.85 throughout this question.

Part I: Reproduce the Analytical Result

(a) For n = 4, calculate P(X = x) computationally for x = 0, 1, 2, 3, 4 using the Binomial probability formula. Display the number of successes x and the corresponding probability P(X = x).

(b) Compare the computational probabilities with the analytical probabilities obtained in Q1. Are the two sets of results the same?

(c) Using Python, determine (i) the maximum probability and (ii) the value of x at which this maximum occurs. Hence, determine computationally the most likely number of successes.

(d) Calculate the total probability:

$$S = \Sigma_{x=0}^{4} P(X = x)$$

Print the value of S. Is the value exactly 1 or very close to 1? Explain your observation.

Part II: Scale the Number of Experiments

Now increase the number of independent experiments while keeping p = 0.85. Repeat the calculation for:

$$n = 10, 20, 50, 100$$

For each value of n:

(e) Calculate P(X = x) for every possible value x = 0, 1, 2, ..., n.

(f) Find computationally the value of x having the maximum probability, the maximum probability, and the ratio x_peak / n, where x_peak represents the most likely number of successes.

(g) Calculate the sum of all possible probabilities:

$$S = \Sigma_{x=0}^{n} P(X = x)$$

Record your results in the following table:

n	x_peak	Maximum Probability	x_peak / n	Sum of All Probabilities
4				
10				
20				
50				
100				

(h) Plot P(X = x) against x for n = 4, 20, 50, and 100. Provide proper axis labels, title, and legend.

(i) Observe the location of the peak as n increases. Compare x_peak / n with p = 0.85. What do you observe as the number of experiments becomes larger? Explain the statistical meaning of your observation.

(j) The theoretical sum of all probabilities is 1. However, a computer may sometimes produce a value such as 0.999999999999 or 1.000000000001. Explain why a computer may produce a value extremely close to, but not exactly equal to, 1.

Q4) Computational Analysis of the Continuous Probability Distribution

Consider again the continuous probability density function from Q2:

$$f(x) = 6x(1 - x), \quad 0 \leq x \leq 1$$

Write a Python program to study this probability distribution computationally.

Part I: Reproduce the Analytical Result

(a) Take $\Delta x = 0.1$ and generate x = 0, 0.1, 0.2, ..., 1.0. Calculate f(x) for every value of x.

(b) Compare the computational values with the analytical values obtained in Q2. Are the results the same?

(c) Find computationally (i) the maximum value of f(x), and (ii) the value of x corresponding to the maximum. Compare the computational result with the analytical maximum obtained in Q2.

(d) Approximate the total probability using:

$$S = \Sigma_i \, f(x_i) \, \Delta x$$

Print the calculated value. Is the result exactly equal to 1?

Part II: Increase the Numerical Resolution

Now investigate what happens when the spacing between consecutive x-values is decreased. Repeat the calculation using:

$$\Delta x = 0.1, 0.05, 0.01, 0.001$$

Keep the range fixed as $0 \leq x \leq 1$. For every value of Δx:

(e) Generate all values of x and calculate f(x).

(f) Determine the number of computational points, the maximum value of f(x), the value x_peak at which the maximum occurs, and the approximate total probability.

$$S = \Sigma_i \, f(x_i) \, \Delta x$$

Record the results in the following table:

Δx	Number of Points	x_peak	Maximum f(x)	Approx. Total Probability
0.1				
0.05				
0.01				
0.001				

(g) Plot f(x) against x for different values of Δx. Observe how the representation of the continuous distribution changes as Δx becomes smaller.

(h) For every value of Δx, calculate both quantities below:

$$S_1 = \Sigma_i \, f(x_i)$$

$$S_2 = \Sigma_i \, f(x_i) \, \Delta x$$

Which of these quantities approaches 1? Explain the reason for the observed behaviour.

(i) As Δx decreases, the number of computational points increases. Discuss the relationship between numerical resolution, numerical accuracy, number of computational points, and computational cost.

(j) Based on your results, explain why numerical integration of a continuous probability density function can be approximated as:

$$\int_a^b f(x) \, dx \approx \Sigma_i \, f(x_i) \, \Delta x$$

Why is multiplication by Δx necessary in the continuous case?

Final Observation

Based on Q1-Q4, briefly explain the main difference between a discrete probability distribution and a continuous probability distribution with respect to the calculation of total probability.